

Experimental investigation of the effects of a compatibilizing agent on the properties of a recycled poly(ethylene terephthalate)/polypropylene blend

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Abstract This study deals with the behavior of a recycled polyethylene terephthalate (PET)/polypropylene (PP) blends. The compatibilizing effect has been investigated to examine the recycling feasibility in industrial production. The compatibilizing efficiency of olefinic copolymers containing epoxy groups for a/polypropylene (PET/PP) blends was examined using scanning electron microscopy (SEM), differential scanning calorimetry (DSC), mechanical testing and rheological one. The effect of ethylene-glycidyl methacrylate (E-GMA, 92/8 wt%, Lotader AX8840) and ethylene-methyl acrylate-glycidyl methacrylate (E-MA-GMA, 68/24/8 wt%, Lotader AX8900) copolymers was investigated. The blends of PET/PP/compatibilizer at compositions 80/15/5, 85/11.25/3.75, 90/7.5/2.5 and 95/3.75/1.25 (wt%) were prepared by melt mixing in a single-screw extruder. Test specimens were prepared by compression moulding at processing temperatures of 250 °C. The incorporation of the compatibilizers has a large effect on the dispersion of the PP phase. Moreover, the copolymer was more efficient than the terpolymer. Especially, E-GMA was found to improve the elongation at break of the blends containing 80 % PET.

Keywords Polyethylene terephthalate · Polypropylene · Polymer blends · Mechanical properties · Morphology · Compatibilizer

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Introduction

Poly(ethylene terephthalate) (PET) is one of the most usually used thermoplastic polyesters. PET is widely used in industrial productions, such as beverage bottles, packaging films and textile fibers, because of its better chemical, physical and mechanical properties. Due to the increasing concern about environmental issues, recycling of PET attracts large importance [1–3]. Nevertheless, during the recycling processes PET thermal and hydrolytic degradations lead to a reduction in the molecular weight of the polymer and, consequently, its viscosity, melt strength and mechanical properties.

Blending polyolefins with engineering plastics is an effective way to obtain materials with specific properties. It can be used in the course of material recycling especially during the melt processing of PET wastes from bottles that generally contain a small percent of olefinic species coming from the cap and when the sorting step is to be avoided [4]. Melt blending is a simple and reliable technique that is suitable for many polymer systems. However, most of the polymers are immiscible. Therefore, blending usually leads to heterogeneous morphologies. The type, shape and size of the morphology that is formed during melt processing, depend on the nature of the polymers (interfacial tension, viscosity and viscosity ratio), on the composition as well as on the processing conditions. In the last decades, many research groups have focused on the control of the morphology and interfaces in immiscible polymer blends to improve their compatibility. The compatibility of immiscible polymers may be improved by adding copolymers, and especially block and graft copolymers with segments capable of specific interactions and/or chemical reactions with the blend constituents or by blending suitable functionalized polymers having specific enhanced interactions and/or chemical reactions. These copolymers, which are called compatibilizers, can be added or formed in situ during melt blending through the interfacial chemical reaction of the polymeric components [5–8]. Usually, the compatibilizer concentrates at interfaces during blending, and plays the role of reducing interfacial tension, preventing coalescence and strengthening interfacial adhesion. Such compatibilization often results in stabilized morphologies with a fine dispersion of the minor phase, which affects the macroscopic properties of the blends. The compatibilization of polymers has been extensively studied through the diminution of the interfacial tension of the polymers in the blend.

Polyethylene terephthalate (PET) and polypropylene (PP) are incompatible due to differences in chemical nature and polarity [9]. Therefore, these physical blends exhibit a two-phase morphology, with large domains and poor adhesion between the phases.

Numerous authors have addressed compatibilization by various copolymers. Bataille et al. [10] reported that the addition of an acrylic acid grafted polypropylene (PP-g-AA) does not produce any significant effect on the particle size or mechanical properties of PET/PP blends. The efficiency in reducing the size of the dispersed phase with PP-g-AA was obtained by Young et al. [11] by pre-blending PP-g-AA with PET to form a graft copolymer of PET and PP. This copolymer is subsequently

added to the PP/PET blend. Cheung et al. [12] used a low molecular weight maleated PP and found poor efficiency in improving the elongation at break though the tensile strength and stiffness of the blends were improved. Though maleic anhydride grafted polypropylene (PP-g-MAH) reacts with PET [13], the efficiency is poor when it is added at low amounts as a compatibilizer in PET/PP blends [14]. The addition of a transesterification catalyst may increase the efficiency of both PP-g-AA and PP-g-MAH [15]. Glycidyl methacrylate grafted polypropylene (PP-g-GMA) is more efficient as an emulsifier though high grafting degree is required; high elongation at break and ductility can be achieved with this compatibilizer [16]. A comparison of the efficiency of various compatibilizers, PP-g-MAH, PP-g-AA and PP-g-GMA on the compatibility of polybutylene terephthalate (PBT)/PP blends was performed by Sun et al. [17, 18] and Hu et al. [19]. The process was a one-step process including free radical grafting of monomers on PP and compatibilization. The authors reported that PP-g-GMA is the most effective compatibilizer while acrylic acid as the active functionality of a compatibilizer has the disadvantage of being kinetically slow when esterified with the terminal OH groups of polyesters. The efficiency of other monomers melt-grafted onto PP such as esters (monomethyl itaconate or dimethyl itaconate) [20], amides (*N,N*-dihydroxyethyl monomaleic amide alone or combined with a tert-butyl phenolic resin) [21] have also been investigated. Pang et al. [22] have performed a comparison of various ester and amide derivatives from the maleic anhydride [*N,N*-dihydroxyethyl monomaleic amide, octodecyl monomaleate, and 2-(*N,N*-dihydroxyethylamino)ethyl monomaleate], only the amide and the ethyl monomaleate combined with a tert-butyl phenolic resin were found to be effective.

Various authors have also addressed the use of compatibilizing copolymers based on other resins than PP. Styrene-*b*-(ethylene-*co*-butylene) copolymers (SEBS) have received much attention either in non-functionalized or functionalized form with various reacting groups. The comparisons have shown that the effect on impact strength is increasing in the order SEBS, SEBS-g-MAH, SEBS-g-GMA [23]. Copolymers such as SEBS-g-MAH have the dual role of reducing interfacial tension and preventing coalescence. Efficiency as emulsifier is not sufficient and coalescence suppression is obtained provided that sufficient coverage of the surface is achieved [24]. The comparison of block copolymers with various architecture and structure, SEBS-g-MAH, SEBS-g-GMA, GMA grafted styrene-*b*-(ethylene-propene) diblock copolymer (SEP-g-GMA), GMA grafted styrene-*b*-(ethylene-propene)-*b*-styrene-*b*-(ethylene-propene) tetrablock copolymer (SEPSEP-g-GMA), have confirmed that GMA is more efficient than MAH and that short styrenic blocks are required [25].

Resins having less compatibility with the possibility of using PP phase. Ethylene-*co*-vinyl acetate (EVA) with or without grafting of maleic anhydride (EVA-g-MAH) are mentioned but with quite low efficiency as regards the improvements in mechanical properties [26]. The comparison of various polymers grafted with maleic anhydride, PP-g-MAH, linear low density polyethylene grafted with maleic anhydride (LLDPE-g-MAH), SEBS-g-MAH, combination of ethylene-propylene copolymer and PP-g-MAH have shown that the latter could be useful in preventing the loss of compatibilizer in the matrix. Therefore, the high level of elongation at

break can be obtained, as for SEBS-*g*-MAH [27]. The compatibilization of PBT-PP by polyethylene grafted with glycidyl methacrylate (E-*g*-GMA) was performed with the addition of a catalyst such as ethyltriphenyl phosphonium bromide to get a significant improvement of the properties [28]. At last, it is worth mentioning that low molecular weight organic compounds such as silane coupling agent (trimethoxy silane) can be used for the treatment of PET flakes. In fact, the alkoxy reacts with the hydroxyl groups of PET during melt extrusion with PP and yields a significant improvement of the mechanical properties [29].

In this work, the compatibilization of PET/PP blends by using a diblock copolymer of ethylene-glycidyl methacrylate (E-GMA) and a triblock copolymer of ethylene-methyl acrylate-glycidyl methacrylate (E-MA-GMA) is studied. The main objective is to relate the extent of compatibilization to morphology, thermal, mechanical and rheological properties. The results were interpreted in terms of variation of morphology, microstructure and crystallinity development in the blend.

Experimental

Materials

The materials investigated were recycled poly(ethylene terephthalate), PET, as the matrix (content higher than 80 %) and polypropylene PP as the dispersed phase. The recycled PET was obtained from soft drink bottles and was supplied by Sorepla (France). The polypropylene used in this work was an Appryl product (commercial grade, PP 3030 FA 1, melt volume flow rate = 0.2 g/10 min at 230 °C, 2.16 kg according to ISO 1133 standard). These polymers were compatibilized by a diblock copolymer of ethylene-glycidyl methacrylate (E-GMA), Lotader AX8840 and a triblock copolymer of ethylene-methyl acrylate-glycidyl methacrylate (E-MA-GMA), Lotader AX8900, both supplied by Arkema. Some typical properties of the compatibilizers are listed in Table 1 according to the manufacturer. Both copolymers had a glycidyl methacrylate content of 8 % and the terpolymer contained 24 wt% of acrylate monomer.

Specimen preparation

PET pellets were dried for 24 h at 120 °C in dehumidifying vacuum prior to processing. The extrusion of the mixtures (PET/PP without and with compatibilizers) was performed on a single-screw extruder type Yvroud HE with a three-zone standard screw ($D = 25$ mm, $L = 28D$). The temperature profile, starting from the feeding zone to the die, was 260, 265, 270 and 260 °C, and the screw speed was maintained at 110 rpm.

Scanning electron microscopy

The morphology of the blends was investigated using a scanning electron microscope SEM Philips XL30. The samples were fractured at liquid nitrogen

Table 1 Properties of the compatibilizers E-GMA and E-MA-GMA (data from the producer)

Copolymer	Reference	Melt Flow Index (g/10 min) ASTM D 1238	Melting point (°C)	Acrylic ester content (wt%)	Glycidyl methacrylate content (wt%)	Young's modulus (MPa)	Tensile strength at break (MPa)	Elongation at break (%)
E-GMA	AX8840	5	105	0	8	104	8	400
E-MA- GMA	AX8900	6	60	24	8	8	4	1100

temperature. The fracture surfaces were observed under an accelerating voltage of 40 kV after gold coating.

Thermal analysis

Differential scanning calorimetry (DSC) measurements were carried out using a Netzsch DSC 204. The weight of the sample was nearly 5 mg and the thermal cycle for the determination of the crystallinity was a temperature ramp from -20 until 280 °C with 10 °C/min heating rate.

Mechanical testing

The specimens for mechanical tests were obtained by compression molding. The useful processing conditions were temperature of 250 °C, above that of the extrusion cycle, in order to prevent further degradation, pressure of 300 kg/cm², time under pressure of 1 min. After melting and forming in the mold, the resulting plates were cooled down to room temperature during 10 min for the crystallization of the components. Tensile tests were carried out using a Lloyd instruments EZ50 machine with a crosshead speed of 1 mm min⁻¹. Five tensile specimens were tested for each formulation.

Rheological testing

The evolution of the viscosities of the PET and the PP in function of the shear rates at temperature of 260 °C were determined by a capillary rheometer (HAAKE).

Results and discussion

Morphology of the blends

Mixing two polymers together generally does not guarantee good interfacial adhesion between the two phases. Proper interfacial bonding is required if the blend is to have good mechanical properties. However, incompatibility between the two polymers usually leads to poor interfacial adhesion. The morphology of the dispersed phase will depend on the degree of mixing, volume fraction and interactions of the two polymers. The mixing of PET and PP with a low amount of PP results in droplets of PP dispersed in the PET matrix. The morphologies of PET/PP blends without compatibilizer are presented in Fig. 1. Without any compatibilizer, two distinct incompatible phases are observed and the PP phase in physical blend is easily identified. As expected, the interpenetration of chains from the two polymers at the interface is poor, thus resulting in poor interfacial adhesion. Figure 1 shows that most of the PP droplets have been extracted from the PET matrix during the fracture process. In addition, the PP phase shows a large distribution of droplet sizes in all cases. The average particle size clearly increases with the PP content due to enhanced coalescence. For the 95/5 composition

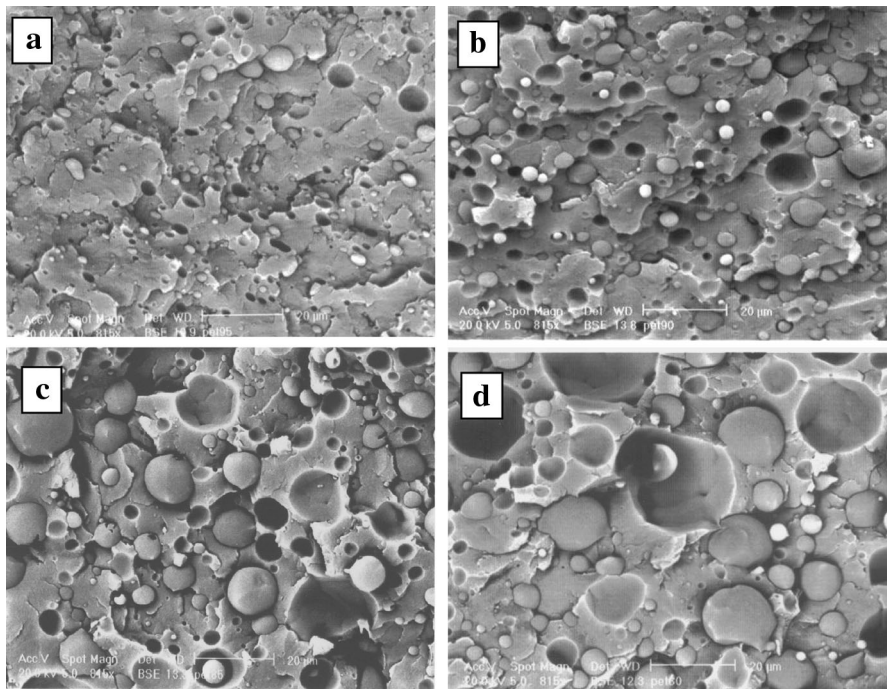


Fig. 1 SEM micrographs of cryo-fractured surface of PET/PP blends prepared without compatibilizer: **a** PET95/PP5, **b** PET90/PP10, **c** PET85/PP15 and **d** PET80/PP20 (X815)

(Fig. 1a), the particle sizes range from 1 to 7 μm with most of the particles having a diameter of 4 μm . At 20 wt% of PP (Fig. 1d). This average reaches 15 μm with droplets as large as 25 μm .

The morphologies of PET/PP blends compatibilized with a diblock copolymer E-GMA (Fig. 2) and with a triblock copolymer E-MA-GMA (Fig. 3) show a finer and more homogeneous distribution of PP spheres in the PET matrix as the result of the improvement in dispersibility generated by the presence of compatibilizers. This clear difference of dispersibility can be attributed to the reaction of epoxy groups of GMA in the compatibilizers and PET. The addition of compatibilizers leads to regular shaped particles and relative uniformly sized PP domains. The compatibilizer has two main roles in the control of morphology of the blend not only by preventing coalescence, but also by the reduction of interfacial tension. In blends prepared by melt mixing, the uniformity of shape and size of the PP particles caused by the addition of compatibilizers is probably the result of the reduction of coalescence because of the steric stabilization role of compatibilizers. Very small amount of copolymer at the interface will provide the necessary effect.

On the contrary, the decrease of the average size is mainly connected to the coverage of the PP particles by the copolymer. In particular, when the size reduces, a larger amount of copolymer is required due to the increase of surface area. Besides, the reduction of the interfacial tension that provides the emulsifying effect

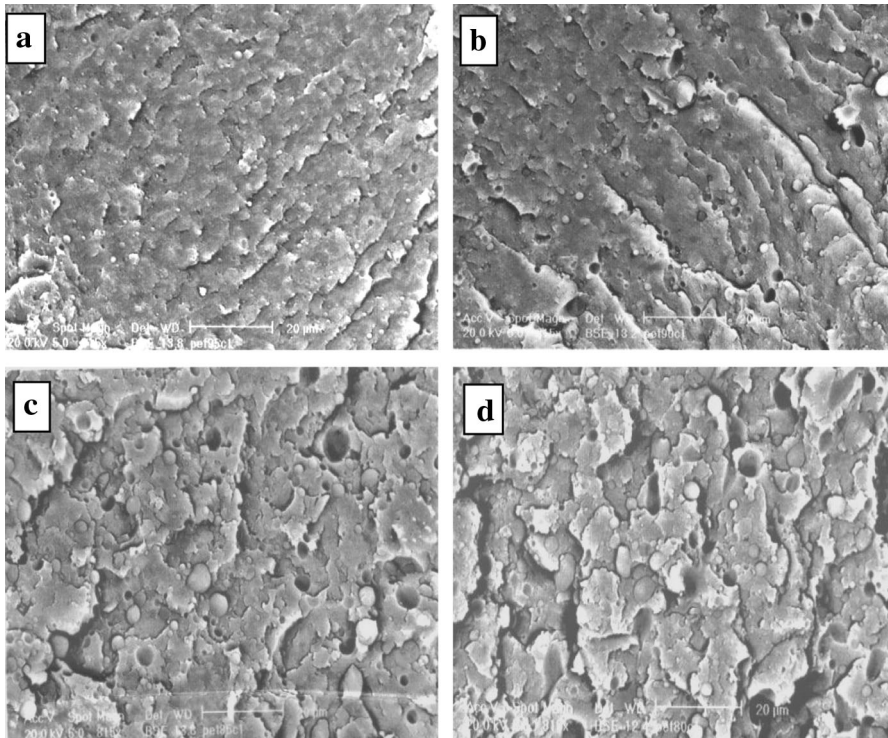


Fig. 2 SEM micrographs of cryo-fractured surface of PET/PP blends prepared with E-GMA: **a** PET95/PP3.75/E-GMA1.25, **b** PET90/PP7.5/E-GMA2.5, **c** PET85/PP11.25/E-GMA3.75 and **d** PET80/PP15/E-GMA5 (X815)

is clearly connected to the extent of the interfacial reaction and to the number of bonds that can be created by unit surface area. It should be noticed that from this point of view, the diblock copolymer of ethylene-glycidyl methacrylate E-GMA produces finer PP domain sizes, and it is more efficient in reducing the interfacial tension than a triblock copolymer of ethylene-methyl acrylate-glycidyl methacrylate, E-MA-GMA. Indeed, comparing the compatibilizing effect of the two different copolymers (E-GMA and E-MA-GMA) in Figs. 2, 3, respectively, it can be observed that for the same composition, the distribution of dispersed phase particles is more homogeneous with lower dimensions of PP particles in PET/PP blends compatibilized with E-GMA. Figure 2 also shows that the presence of E-GMA leads to a large reduction of the empty cavities left by extracted PP nodules. For the 95/5 blends, individual droplets can hardly be identified and become rather undistinguishable from the matrix when the E-GMA compatibilizer is used, the average size being less than 1 μm with very few particles detached from the PET. For the E-MA-GMA compatibilizer, the size is 2 μm with droplets as large as 4 μm . The increase of the average size with the PP amount is visible, and the difference between the effects of the two compatibilizers is strengthened within the 80/20 blends. In this case, the maximum size is 4 μm for the E-GMA copolymer while it

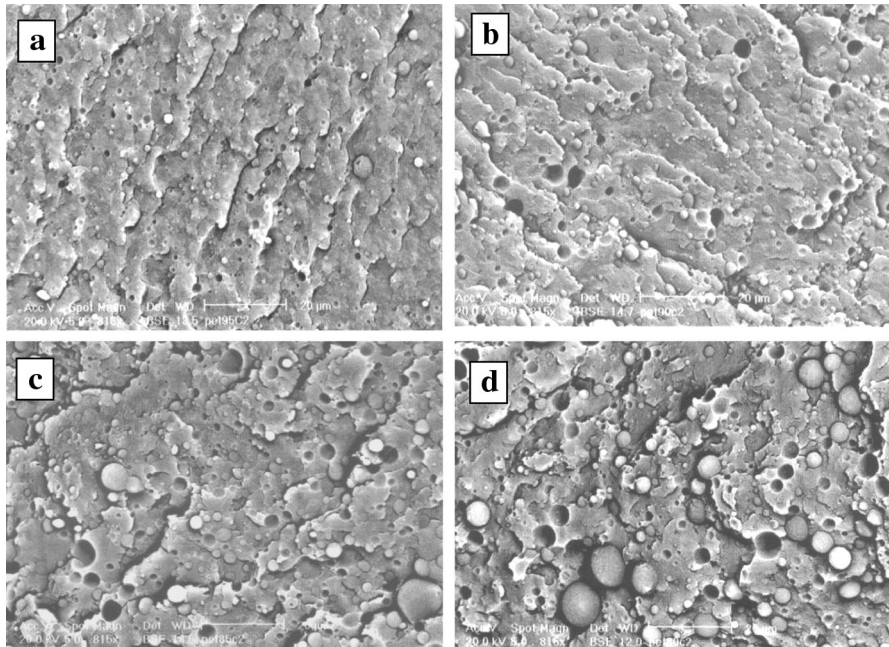


Fig. 3 SEM micrographs of cryo-fractured surface of PET/PP blends prepared with E-MA-GMA: **a** PET95/PP3.75/E-MA-GMA1.25, **b** PET90/PP7.5/E-MA-GMA2.5, **c** PET85/PP11.25/E-MA-GMA3.75 and **d** PET80/PP15/E-MA-GMA5 (X815)

remains much higher for the terpolymer. The effect of the compatibilizers on the droplet size is summarized in Fig. 3. The large difference is due to the acrylate ester monomer that is neither miscible with PP nor reactive with PET. For the same amount of copolymer, though the wt% of reactive groups is the same for both copolymers, the efficiency of the reaction at the interface between the PET and PP is limited for the E-MA-GMA copolymer because of the steric hindrance of the acrylate ester block. Therefore, the number of connections between the phases per unit area of interface is lower for the E-MA-GMA copolymer. This means that part of the copolymer will be dispersed in the PET phase as micelles of copolymer explaining the small droplets that are clearly visible in Fig. 2. The weakness of the interfacial bonding can also be at the origin of the poor adhesion that is noticed through the detachment of PP droplets during the fracture for the E-MA-GMA copolymer. In addition, there might be some effects of the length of the ethylene blocks that have different miscibility with the PP matrix. Therefore, the presence of E-GMA in the PET/PP blends will result in a larger decrease in interfacial tension, better dispersion of the minor phase and increased adhesion and resistance to debonding.

Similar results was observed by Catelli de Souza [3] where a comparative study of recycled PET/PP-EP and recycled PET/PP blend showed that for all compatibilized blends present droplet dispersion morphology. The dispersed phase diameters decreased with the compatibilizer concentration. In addition, Razak

et al. [30] found that the addition of PP-*g*-MAH and SEBS-*g*-MAH compatibilizers into the blends promote a better dispersion of PP into PET matrix.

Thermal properties

In addition to changes in morphology, some effects of the copolymers on the thermal properties of the blends can also be expected. The DSC results for neat PP, recycled PET and blends without compatibilizers are shown in Fig. 4 and Table 2. No significant difference can be noted apart from an increase of the glass transition temperature of PET for the largest amount of PP and a slight decrease of the melting temperature of PET. This is probably due to the physical restriction of the PP domain to the mobility of PET chains.

For the blends of PET, PP and copolymer, Figs. 5, 6, 7 and Table 2 show a decrease of the heat of fusion of the polypropylene phase. This decrease indicates a lower amount of crystallinity showing that the crystallization of the PP phase is slightly hindered by the presence of the copolymers. Simultaneously, a slight decrease of the PP melting point is observed, which could indicate defects in the crystalline phase of PP.

The primary reason could be the miscibility of the copolymer with the PP phase. However, the DSC results for blends of PP with compatibilizer in a 3:1 ratio do not show significant differences between the melting point and heat of fusion of the PP phase (Table 2). A second reason could be related to the droplet morphology that generates a fractionated crystallization process. Indeed, for small droplets, the number of nuclei in PP can be lower than the number of particles and in this case the crystallization of PP in non-nucleated particles proceeds in a homogeneous way, giving rise to distinct phases.

The situation is clearly different for the two copolymers. The E-GMA copolymer induces stronger effects than the E-MA-GMA copolymer. A decay of the heat of fusion of the PP phase by a factor of 15 % is observed for the E-GMA copolymer,

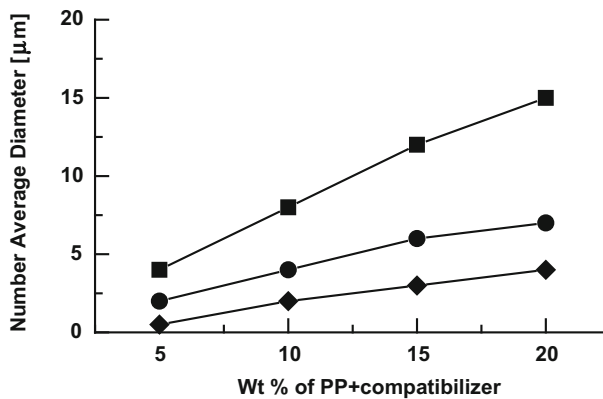
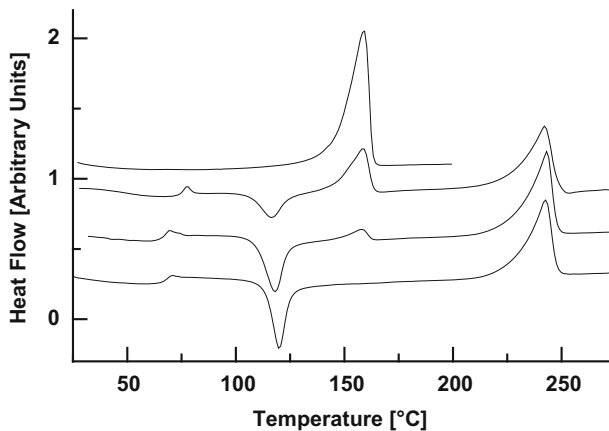


Fig. 4 Number average diameter of dispersed droplets: *square* no compatibilizer, *diamond* E-GMA compatibilizer, *circle* E-MA-GMA compatibilizer. For compatibilized blends, the ratio compatibilizer/(PP + compatibilizer) is 0.25

Table 2 Thermal properties of the PET/PP blends

PET	PP	E-GMA	E-MA-GMA	$T_{\text{g}}(\text{PET})$ (°C)	$T_{\text{c}}(\text{PET})$ (°C)	$T_{\text{f}}(\text{PET})$ (°C)	$T_{\text{f}}(\text{PP})$ (°C)	$\Delta H_{\text{f}}(\text{PP})$ (J/g)	$\Delta H_{\text{c}}(\text{PET})$ (J/g)	$\Delta H_{\text{f}}(\text{PET})$ (J/g)
100	0	0	0	75.7	127.7	250.7	–	–	27.2	55.4
95	5	0	0	75.1	126.2	251.2	166.2	119.7	25.5	53.9
80	20	0	0	81.1	127.6	250.1	166.1	124.1	23.3	55.1
95	3.75	1.25	0	76.0	124.0	250.0	164.5	104.8	20	53.4
80	15	5	0	80.3	124.5	248.4	165.5	103.7	23.8	52.1
95	3.75	0	1.25	76.2	126.1	250.2	165.2	116.0	23.6	51.2
80	15	0	5	80.9	126.9	249.0	165.9	113.4	23.3	53.7
0	100	0	0	–	–	–	167.3	122.0	–	–
0	75	25	0	–	–	–	167.4	120.1	–	–
0	75	0	25	–	–	–	168.2	121.3	–	–

**Fig. 5** DSC thermograms for the non-compatible PET/PP blends and the neat components (*from the top* PP, PET80/PP20PET, PET95/PP5, PP)

while it is only 5 % for the terpolymer. However, the assumptions of miscibility or fractionated crystallization can hardly be distinguished from these observations since both of them can be invoked in the case of the E-GMA copolymer. In comparison with the E-MA-GMA copolymer, the E-GMA is more miscible with PP but it also gives rise to a better dispersion as stated previously from the morphology. The lower crystallization temperature of the PET phase, in the presence of the E-GMA copolymer, could be an indication of a slight nucleating effect of the highly dispersed PP phase. Indeed, in this case the diameter of the dispersed particle is submicronic.

Mechanical properties

Table 3 show the changes in the tensile modulus, tensile strength at break and elongation at break for the various components and blends. At the low amounts of PP, the addition of

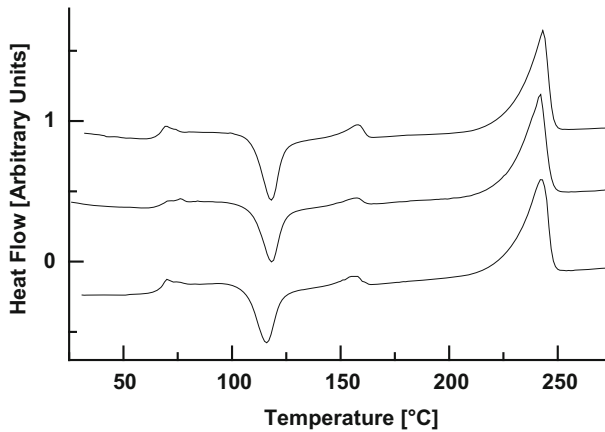


Fig. 6 DSC thermograms for PET/PP blends containing 95 % PET (*from the top* PET95/PP5, PET95/PP3.75/E-MA-GMA1.25, PET95/PP3.75/E-GMA1.25)

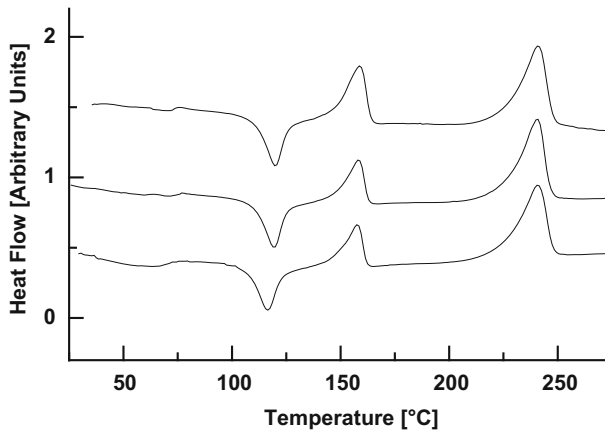


Fig. 7 DSC thermograms for PET/PP blends containing 80 % PET (*from the top* PET80/PP20, PET80/PP15/E-MA-GMA5, PET80/PP15/E-GMA5)

PP to the blend induces a large decrease of the tensile modulus. The decrease is not related to a lower crystallinity of the PET phase since for all the samples, the degree of crystallinity of the PP phase at room temperature is nearly 30 % (Table 2). The modulus of the blends is generally lower than the value which could be predicted by a simple linear blending law, indicating poor adhesion between the components. The effect of the compatibilizer cannot be assessed by measurements at low strains because the addition of the copolymer (E-GMA or E-MA-GMA) itself leads to a larger decrease of the tensile modulus with regard to all compositions of the blends without emulsifying agent. This is explained by the elastomeric behavior of these copolymers whose modulus is significantly lower than that of neat PP. Nevertheless, for a large amount of PP, the E-GMA compatibilizer is more efficient in maintaining the highest value of the modulus.

Table 3 Mechanical properties of PET/PP blends

PET	PP	E-GMA	E-MA-GMA	Modulus (MPa)	Tensile strength (MPa)	Elongation at break (%)
100	0	0	0	860 ± 50	37 ± 5	6 ± 3
95	5	0	0	710 ± 60	27 ± 4	8 ± 4
90	10	0	0	690 ± 50	21 ± 4	9 ± 4
85	15	0	0	680 ± 50	19 ± 3	12 ± 5
80	20	0	0	520 ± 50	16 ± 2	13 ± 6
0	100	0	0	200 ± 50	12 ± 5	120 ± 30
95	3.75	1.25	0	370 ± 60	27 ± 5	16 ± 4
90	7.5	2.5	0	400 ± 90	22 ± 8	15 ± 4
85	11.25	3.75	0	410 ± 80	22 ± 6	17 ± 3
80	15	5	0	470 ± 40	23 ± 5	25 ± 2
95	3.75	0	1.25	420 ± 50	31 ± 4	12 ± 3
90	7.5	0	2.5	370 ± 60	27 ± 4	14 ± 3
85	11.25	0	3.75	400 ± 40	26 ± 3	13 ± 2
80	15	0	5	360 ± 40	20 ± 4	13 ± 3
95	0	5	0	410 ± 50	35 ± 7	6 ± 3
95	0	0	5	470 ± 30	36 ± 3	9 ± 2

As usually observed, the mechanical properties at break are more susceptible to differences in the compatibilization efficiency. The tensile strength of neat PP and recycled PET were found to be 12 ± 5 and 37 ± 5 MPa, respectively. Adding PP to PET induces a decrease of the tensile strength but the decrease in tensile strength on adding 5 and 20 % PP is much larger than expected because of poor interfacial adhesion between PET and PP. For the PET/PP blends, the addition of E-GMA or E-MA-GMA gives a significant improvement in the tensile strength. In particular, the comparison of blends containing 95 % of PET with 5 % of compatibilizer shows that the tensile strength of the blends remains at the same level as that of pure PET, while it decreases when pure PP is added because of poor adhesion in the latter. The tensile strength of blends with compatibilizer is closer to that calculated by a simple blending law. These data indicate improved adhesion between the two phases, which is a necessary condition for improving the mechanical strength. The favorable effect of the E-GMA compatibilizer seems to emerge, especially at high amount of PP.

Akbari et al. [31] observed that compatibilizing effects for PET/PP blends are dependent on the quantity loading of PP-*g*-MAH into the blends. Their result shows that the finest PET domain was at 10 wt% of PP-*g*-MAH. The reaction of ester groups of MAH with the PET/PP blends during melt mixing, increased the tensile strength, tensile modulus, and elongation at break of the blends.

The effect of the PP on the elongation at break of PET is clearly displayed in Table 3. It should be recalled that the crystallinity of PET is nearly the same for all blends. Adding PP to PET yields higher elongation at break. Indeed, for all blends with a PET matrix in this study, fracture takes place by shear yielding and crazing.

Therefore, the addition of a ductile PP phase has an effect on the fracture termination. However, toughening is also related to the increase of the dissipation mechanism that is obtained for fine morphologies provided that the adhesion between the phases is sufficient. For these reasons, adding a copolymer gives way to higher values of elongation at break since the size of the dispersed phase is smaller. The E-GMA copolymer is more efficient in the whole composition range. The results of tensile strength indicate that although the adhesion is strong enough for both compatibilizers at room temperature, the finest morphology of blends with the E-GMA copolymer has the most favorable effect on elongation at break and overall toughening mechanism. Akkapeddi et al. [32] have obtained similar results on PET/hdPE blends with identical weight fractions of both polymers.

Rheological properties

The results are shown in Fig. 8. The rheological characteristics of the neat components, namely, PP and PET. The rheological behavior of PET/PP blends compatibilized with both the diblock copolymer (E-GMA) shows that the triblock copolymer (E-EA-GMA) is typically pseudoplastic. Like their non-compatibilized counterparts, flow curves obtained converge to the high shear rates, while the differences in viscosities of various compatibilized blends are more significant at low shear rates. Similarly, over the entire studied shear range, the viscosity of the compatibilized blends is located below those of the two constituents and pure PP/PET. It is also noted that the more the percentage of PP in the blend increases, the more the viscosity decreases. This finding is also valid for non-compatibilized blends, which is due partly to the low viscosity of the PP phase relative to the matrix and the other to a more stable and fine morphology. Figure 9 shows the variation of the viscosity of various blends PET/PP compatibilized and non-compatibilized according to the percentage of PP in the blend at an intermediate shear rate (100 s^{-1}).

It is found that the presence of the interfacial agents has the effect of reducing the viscosity of the PET/PP mixtures over the entire studied composition range. This

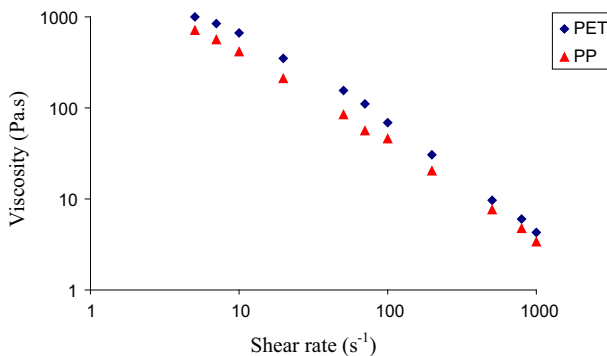


Fig. 8 Evolution of the viscosity of PET and PP versus shear rate ($T = 260 \text{ }^{\circ}\text{C}$)

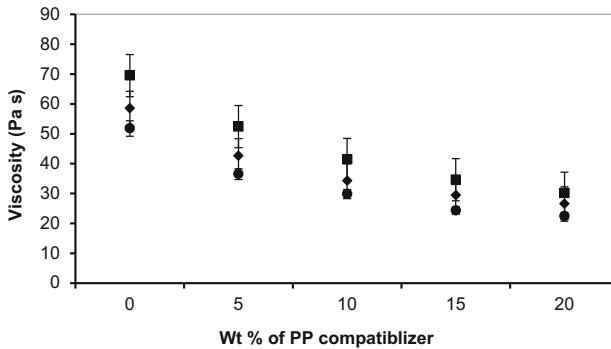


Fig. 9 Viscosity of PET/PP blends: *square* no compatibilizer, *diamond* E-GMA compatibilizer, *circle* E-MA-GMA compatibilizer. For compatibilized blends, the ratio compatibilizer/(PP + compatibilizer) is 0.25

reduction in viscosity is attributed to a nodular morphology with dimensions of the lower dispersed phase in the presence of the compatibilizer. The decrease in viscosity values for compatibilized blends is also attributed to the lower viscosity of the third compatibilizing body. It is worthy to note that the decrease in the viscosity of the compatibilized, and not all compatibilized blends, is particularly favored, and that the rate of the polypropylene in the mixture is important. Moreover, blends compatibilized with the triblock copolymer have lower viscosities as compared to blends compatibilized with the diblock copolymer. This is attributed to a better transmission of stresses between matrix and dispersed phase in the presence of E-GMA copolymer. However, in the presence of the triblock copolymer E-EA-GMA, the density of the interactive forces between the two phases is less, and therefore the viscosity is lower. On the other hand, the triblock E-EA-GMA has a lower viscosity compared with that of diblock E-GMA. It is noted that the more the percentage of PP in the blend increases, the more the effect of the compatibilizer on the decrease of the viscosity of the mixture diminishes. This is due to the competition of the two factors, namely the reduction of the size of the dispersed phase and the increase in the percentage of PP in the blend. Indeed, with the increase in the percentage of PP, the size of particles dispersed in the compatibilized blends is significant, resulting in a less effect on reducing the viscosity of the mixtures.

It is noted over the entire shear rate, the addition of E-GMA copolymer decreases the viscosity of the blends PP20. However, the effect of E-EA-GMA terpolymer on the decrease in the viscosity, at the same content in PP is more important than that of E-GMA copolymer (Fig. 10). This may result from the difference between these two types of copolymers at the interface. Indeed, the diblock copolymer compatibilizer to the interface is organized in a monolayer around the particles while being miscible with PP phase. Nevertheless, the triblock is positioned randomly and therefore, as shown in the morphological study, the triblock copolymer of the layer at the interface tends to thicken.

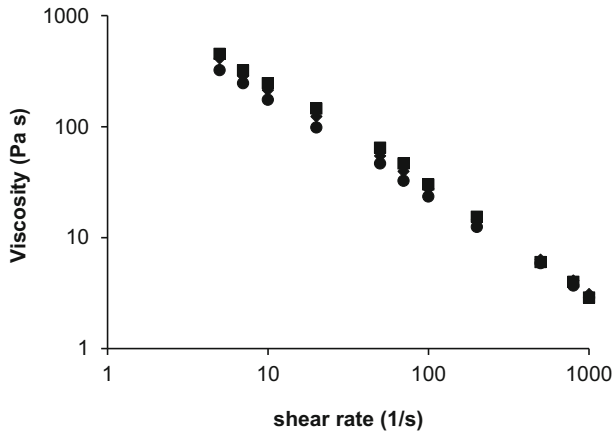


Fig. 10 Evolution of the viscosity of PET/PP blends versus shear rate ($T = 200 \text{ } ^\circ\text{C}$); *square* no compatibilizer, *diamond* E-GMA compatibilizer, *circle* E-MA-GMA compatibilizer. For compatibilized blends, the ratio compatibilizer/(PP + compatibilizer) is 0.25

Conclusion

The effect of E-GMA and E-MA-GMA compatibilizers on the morphology as well as the thermal, mechanical and rheological properties of PET/PP blends containing low amounts of PP (less than 80 %) was investigated.

1. Compatibilized PET/E-GMA/PP and PET/E-MA-GMA/PP blends with different compositions have generally shown a consistent decrease of the mean size of the dispersed phase and improved interfacial adhesion, as compared to non-compatibilized PET/HDPE blends. The highest compatibilizing effectiveness was observed for E-GMA copolymer that causes the highest decrease of the average size for dispersed PP particles. The steric hindrance of the acrylate ester block in the other copolymer has a disruptive effect for the coverage of the dispersed particles and the number of bonds that are created at the interface.
2. The DSC measurements have also indicated a stronger disturbance of the crystallization of the PP phase when using the E-GMA copolymer. Although better miscibility with PP could be expected with this copolymer, the effect is probably related to the size of the dispersed phase.
3. The mechanical properties of PET/PP blends were markedly affected by E-GMA, located at the matrix/dispersed phase interface. Nevertheless, the addition of E-GMA improves the elongation at break of the PET/PP blends and provides interesting toughening. On the other hand, E-MA-GMA copolymer yields sufficient adhesion at room temperature, but insufficient emulsifying effect, leading to a coarser morphology and a lower toughening mechanism.
4. The rheological study has demonstrated that the presence of a third body compatibilizer facilitates the transformation of PET/PP blends and results in

better control of the morphology of the system on which will depend the final properties of the material.

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References

1. Nait-Ali LK, Colin X, Bergeret A (2011) Kinetic analysis and modelling of PET macromolecular changes during its mechanical recycling by extrusion. *Polymer* 96:236–246
2. Makkama S, Harnnarongchai W (2015) Rheological and mechanical properties of recycled PET modified by reactive extrusion. *Energy Procedia* 56:547–553
3. Martinelli Catelli de Souza A, Barbosa Caldeira C (2015) An investigation on recycled PET/PP and recycled PET/PP-EP compatibilized blends: rheological, morphological, and mechanical properties. *J Appl Polym* 132:41892–41900
4. Itim B, Philip M (2015) Effect of multiple extrusions and influence of PP contamination on the thermal characteristics of bottle grade recycled PET. *Polym Degrad Stab* 117:84–89
5. Fuchs C, Bhattacharyya D (2006) S. Fakirov (2006) Microfibril reinforced polymer–polymer composites: application of Tsai-Hill equation to PP/PET composites. *Compos Sci Technol* 66:3161–3171
6. Villalobos M, Awojulu A, Greeley G, Turco G, Deeter G (2006) Oligomeric chain extenders for economic reprocessing and recycling of condensation plastics. *Energy* 31:3227–3234
7. Canetti CD, Bertini F (2007) Supermolecular structure and thermal properties of poly(ethylene terephthalate)/lignin composites. *Compos Sci Technol* 67:3151–3157
8. Friedrich K, Evstatiev M, Fakirov S, Evstatiev O, Ishii M, Harrass M (2005) Microfibrillar reinforced composites from PET/PP blends: processing, morphology and mechanical properties. *Compos Sci Technol* 65:107–116
9. Rudin A, Loucks DA, Goldwasser JM (1980) Oriented monofilaments from blends of poly(ethylene terephthalate) and polypropylene. *Polym Eng Sci* 20:741–746
10. Bataille P, Boisse S, Schreiber HP (1987) Mechanical properties and permeability of polypropylene and poly(ethylene terephthalate) mixtures. *Polym Eng Sci* 27:622
11. Lepers Jean-Christophe, Favis Basil D, Lacroix Christophe (1999) The influence of partial emulsification on coalescence suppression and interfacial tension reduction in PP/PET blends. *J Polym Sci* 37(9):939–951
12. Cheung MK, Chan D (1997) Mechanical and rheological properties of poly(ethylene terephthalate)/polypropylene blend. *Polym Int* 43(3):281–287
13. Yoon KH, Lee HW, Park OO (1998) Properties of poly(ethylene terephthalate) and maleic anhydride-grafted polypropylene blends by reactive processing. *J Appl Polym Sci* 70:389
14. Oromichie A, Mamizadeh A (2004) Recycling PET beverage bottles and improving properties. *Polym Int* 53:728
15. Xanthos M, Young MW, Biesenberger JA (1990) Polypropylene/polyethylene terephthalate blends compatibilized through functionalization. *Polym Eng Sci* 30:355
16. Champagne MF, Huneault MA, Roux C, Peyrel W (1999) Reactive compatibilization of polypropylene/polyethylene terephthalate blends. *Polym Eng Sci* 39:976
17. Sun YJ, Hu GH, Lambla M (1995) Free radical grafting of glycidyl methacrylate onto polypropylene in a co-rotating twin screw extruder. *J Appl Polym Sci* 57:1043
18. Sun YJ, Hu GH, Kotlar HK, Lambla M (1996) In situ compatibilization of polypropylene and poly(butylene terephthalate) polymer blends by one-step reactive extrusion. *Polymer* 37:4119
19. Hu GH, Sun YJ, Lambla M (1996) Effects of processing parameters on the in situ compatibilization of polypropylene and poly(butylene terephthalate) blends by one-step reactive extrusion. *J Appl Polym Sci* 61:1039–1047
20. Yazdani-Pedram M, Vega H, Retuert J, Quijada R (2003) Compatibilizers based on polypropylene grafted with itaconic acid derivatives. Effect on polypropylene/polyethylene terephthalate blends. *Polym Eng Sci* 43:960

21. Pang YX, Jia DM, Hu HJ, Hourston DJ, Song M (1999) A quantitative estimation of the extent of compatibilization in heterogeneous polymer blends using their heat capacity increment at the glass transition. *J Appl Polym Sci* 74:2868
22. Pang YX, Jia DM, Hu HJ, Hourston DJ, Song M (2000) Effects of a compatibilizing agent on the morphology, interface and mechanical behaviour of polypropylene/poly(ethylene terephthalate) blends. *Polymer* 41:357
23. Heino M, Kirjava J, Hietaoja P, Seppala JC (1997) Compatibilization of polyethylene terephthalate/polypropylene blends with styrene-ethylene/butylene-styrene (SEBS) block copolymers. *J Appl Polym Sci* 65:241
24. Leclair A, Favis Basil D (1996) Role of interfacial contact in immiscible binary polymer blends and its influence on mechanical properties. *Polymer* 37(21):4723–4728
25. Pracella M, Chionna D, Pawlak A, Galeski A (2005) Reactive mixing of PET and PET/PP blends with glycidyl methacrylate-modified styrene-*b*-(ethylene-*co*-olefin) block copolymers. *J Appl Polym Sci* 98(5):2201–2211
26. Yong L, Qinglin W, Clemons Craig M, Guo Weihong (2009) Phase structure and properties of poly(ethylene terephthalate)/high-density polyethylene based on recycled materials. *J Appl Polym Sci* 113:1700–1719
27. Papadopoulos CP, Kalfoglou NK (2000) Comparison of compatibilizer effectiveness for PET/PP blends: their mechanical, thermal and morphology characterization. *Polymer* 41:2543–2555
28. Tsai CH, Chang FC (1996) Polymer blends of PBT and PP compatibilized by ethyleneco-glycidyl methacrylate copolymers. *J Appl Polym Sci* 61:321–332
29. Oyman ZO, Tinçer T (2003) Melt blending of poly(ethylene terephthalate) with polypropylene in the presence of silane coupling agent. *J Appl Polym Sci* 89:1039
30. Abdul Razak NC, Inuwa IM, Hassan A, Samsudin SA (2013) Effects of ompatibilizers on mechanical properties of PET/PP blend. *Compos Interfaces* 20:507–515
31. Akbari M, Zadhoush A, Haghighat M (2006) PET/PP blending by using PP-*g*-MA synthesized by solid phase. *J Appl Polym Sci*. 104:3986–3993
32. Retolaza A, Eguiazábal JI, Nazábal J (2003) Poly (ethylene-*co*-methacrylic acid)–lithium ionomer as a compatibilizer for poly-(ethylene terephthalate)/linear low-density polyethylene blends. *J Appl Polym Sci* 87:1322–1328